A weighted-Hardy counterexample to the reciprocal and essential-spectrum conjectures

Codex agent lane 11

July 21, 2026

Status. Candidate counterexample likely valid, subject to human review. One weighted Hardy Hilbert space refutes the essential-spectrum equality and both general reciprocal conjectures on PDF pages 26–27 of arXiv:2211.12236.

Source questions

Cao, He, and Li ask whether, for a multiplier ψ on a Banach space $B(\Omega)$ of holomorphic functions whose normalized evaluation functionals converge weak-star to zero at the boundary,

$$\bigcap_{K \Subset \Omega} \overline{\psi(\Omega \setminus K)} = \sigma_e(M_\psi). \tag{1}$$

They also propose that $\psi \in B(\Omega)$ and $\inf_\Omega |\psi| > 0$ should imply $1/\psi \in B(\Omega)$, and repeat the conjecture under the stronger assumption $\psi \in \mathcal{M}(B(\Omega))$ [1, Section 6, pp. 26–27].

6. Further Remarks

According to Theorem 5.3, if $M_\psi C_\varphi$ is a Fredholm operator on $B(\Omega)$, then C_φ is also a Fredholm operator, so $\varphi \in \mathrm{Aut}\,(\Omega)$. However, it is difficult to see the characteristics of the symbol ψ according to the Fredholmness of M_ψ. It can be seen from Lemma 5.1 that if M_ψ is bounded on$B(\Omega)$, then $\psi \in H^\infty(\Omega)$, the algebra of bounded holomorphic functions on Ω, and $\overline{\psi(\Omega)} \subset \sigma(M_\psi)$, the spectrum of M_ψ. But we may wonder whether $\overline{\psi(\Omega)} = \sigma(M_\psi)$ or not. If

$$k_z = \frac{K_z}{\|K_z\|} \xrightarrow{weak^*} 0 \quad \text{as} \quad z \to \partial\Omega,$$

then we have

$$\bigcap_{K \subset \Omega} \overline{\psi(\Omega - K)} \subset \sigma_e(M_\psi),$$

where $\sigma_e(M_\psi)$ is the essential spectrum of M_ψ, and K is the compact subset of Ω. But we do not know whether there is

$$\bigcap_{K \subset \Omega} \overline{\psi(\Omega - K)} = \sigma_e(M_\psi).$$

This problem involves another interesting problem, namely the reciprocal problem, which was firstly raised by K.H. Zhu in [37] for the weighted Bergman space. He also proposed the following conjecture

Conjecture 1 ([37]). Let $0 < p < \infty$ and α be real. If $f \in A^p_\alpha$ and there exists a constant $c > 0$ such that $|f(z)| \geq c$ for all $z \in \mathbb{B}_n$, then $1/f \in A^p_\alpha$.

The case when α is somewhere in the middle is difficult. We naturally propose a similar conjecture for $B(\Omega)$:

Conjecture 2. Let $B(\Omega)$ be the Banach space of holomorphic functions on a domain Ω. If $\psi \in B(\Omega)$ and there exists a constant $c > 0$ such that $|\psi(z)| \geq c$ for all $z \in \Omega$, then $1/\psi \in B(\Omega)$.

In [9], the authors proved that if $f \in \mathcal{M}(A^2_\alpha)$, the algebra of multipliers on A^2_α (corresponding Hardy–Sobolev space), then $\overline{f(\mathbb{B}_n)} = \sigma(M_f)$. Furthermore, if there exists a constant $c > 0$ such that $|f(z)| \geq c$ for all $z \in \mathbb{B}_n$, then $1/f \in A^2_\alpha$. Thus there is a weaker conjecture

Conjecture 3. Let $B(\Omega)$ be the Banach space of holomorphic functions on a domain Ω. If $\psi \in \mathcal{M}(B(\Omega))$ and there exists a constant $c > 0$ such that $|\psi(z)| \geq c$ for all $z \in \Omega$, then $1/\psi \in B(\Omega)$.

Obviously, the conjecture is equivalent to $\overline{\psi(\Omega)} = \sigma(M_\psi)$.

The first two broad questions in the paper's introduction are answered in the same paper. The statements above are the live final-section questions treated here.

Definitions

For positive weights $\beta = (\beta_n)_{n\geq 0}$, the weighted Hardy Hilbert space is

$$H^2(\beta) = \left\{ f(z) = \sum_{n\geq 0} c_n z^n : \|f\|_\beta^2 := \sum_{n\geq 0} |c_n|^2 \beta_n^2 < \infty \right\}.$$

The usual requirement $\liminf_n \beta_n^{1/n} \geq 1$ makes every element holomorphic on $\mathbb{D}$; polynomials are dense and the monomials are orthogonal [2]. We write K_z for the reproducing kernel and identify it with the corresponding evaluation functional. The essential spectrum is the Fredholm essential spectrum:

$$\sigma_e(T) = \{\lambda : T - \lambda I \text{ is not Fredholm}\}.$$

Proof intuition

The weights will form longer and longer mountains. From each valley they rise by a factor 2 at every step and later descend by a factor 1/2 until returning to 1. Multiplication by z is therefore a weighted shift. Near a valley it looks for a long interval like a bilateral shift with weights 1/2 on the left and 2 on the right. Its adjoint has a localized approximate eigenvector for every scalar strictly between those two weights; we use $a = 3/2$.

The repeated valleys force $\|K_z\| \to \infty$ at the unit circle, which gives the normalized-kernel hypothesis. The increasingly high peaks make the Taylor series of $1/(a - z)$ fail the weighted square-summability test. Thus the same geometry supplies all three failures.

Counterexample theorem

Theorem 1. *There is a weighted Hardy Hilbert space $B = H^2(\beta)$ on $\mathbb{D}$ such that*

$$\frac{1}{2} \leq \frac{\beta_{n+1}}{\beta_n} \leq 2 \qquad (n \geq 0)$$

and, for $\psi(z) = \frac{3}{2} - z$, all of the following hold.

1. *Polynomials are dense in B, point evaluations are bounded, and $K_z/\|K_z\| \to 0$ weakly (equivalently weak-star in B^*) as $|z| \to 1$.*
2. *$\psi \in \mathcal{M}(B) \cap B$ and $|\psi(z)| \geq 1/2$ on $\mathbb{D}$, but $1/\psi \notin B$.*
3. $$0 \in \sigma_e(M_\psi) \quad \textit{while} \quad \bigcap_{K \Subset \mathbb{D}} \overline{\psi(\mathbb{D} \setminus K)} = \frac{3}{2} - \mathbb{T},$$

   and the circle on the right does not contain zero.

Consequently, (1) *can be a strict inclusion, both reciprocal conjectures are false as stated, and even $\overline{\psi(\mathbb{D})} = \sigma(M_\psi)$ can fail.*

Proof. Set $N_1 = 0$. Recursively, after N_m has been chosen, put

$$L_m = (N_m + 1)^2, \qquad N_{m+1} = N_m + 2L_m.$$

Define one mountain on $[N_m, N_{m+1}]$ by

$$\begin{aligned} \beta_{N_m+j} &= 2^j && (0 \le j \le L_m), \\ \beta_{N_m+L_m+j} &= 2^{L_m-j} && (1 \le j \le L_m). \end{aligned}$$

The definitions agree at consecutive valleys. Thus $\beta_{N_m} = 1$, $\beta_n \ge 1$, and every consecutive ratio is either 2 or 1/2. In particular, $\liminf \beta_n^{1/n} = 1$, and

$$\sum_{n\ge 0} \frac{|z|^{2n}}{\beta_n^2} \le \frac{1}{1-|z|^2} \qquad (z \in \mathbb{D}).$$

Hence $B = H^2(\beta)$ is a reproducing-kernel Hilbert space of holomorphic functions, and polynomials are dense.

Its kernel norm is

$$\|K_z\|^2 = \sum_{n\ge 0} \frac{|z|^{2n}}{\beta_n^2}.$$

Because $\beta_{N_m} = 1$ for every m, monotone convergence gives $\|K_z\| \to \infty$ as $|z| \to 1$. If p is a polynomial, then

$$\left\langle p, \frac{K_z}{\|K_z\|} \right\rangle = \frac{p(z)}{\|K_z\|} \longrightarrow 0.$$

Density of the polynomials and the unit norm of the normalized kernels extend this to every element of B. Since B is Hilbert and hence reflexive, this is also the weak-star convergence of the evaluation functionals required by the source.

Let $e_n = z^n/\beta_n$. Then (e_n) is an orthonormal basis and

$$S := M_z, \qquad Se_n = w_n e_{n+1}, \qquad w_n = \frac{\beta_{n+1}}{\beta_n} \in \{1/2, 2\}.$$

Thus S is bounded and $M_\psi = \frac{3}{2}I - S$ is a bounded multiplier. Also ψ is a polynomial and $|\psi(z)| \ge \frac{3}{2} - |z| > \frac{1}{2}$ on $\mathbb{D}$.

We next show that $a = 3/2$ lies in the Fredholm essential spectrum of S. For $m \ge 3$, choose integers $q_m \to \infty$ so slowly that

$$q_m \le \frac{1}{3} \min\{L_{m-1}, L_m\}.$$

Around the valley N_m, define

$$y_m = \sum_{j=0}^{q_m} \left(\frac{a}{2}\right)^j e_{N_m+j} + \sum_{j=1}^{q_m} \left(\frac{1}{2a}\right)^j e_{N_m-j}, \qquad x_m = \frac{y_m}{\|y_m\|}.$$

The supports can be chosen disjoint, so (x_m) is an orthonormal sequence. On the right of the valley the weights are 2, while on the left they are 1/2. The coefficient recurrence $w_n c_{n+1} = a c_n$ is therefore exact at every interior support index. Only the two endpoints contribute, and

$$\|(S^* - aI)y_m\|^2 = \frac{1}{4}\left(\frac{1}{2a}\right)^{2q_m} + a^2 \left(\frac{a}{2}\right)^{2q_m} \longrightarrow 0.$$

Moreover $1 \le \|y_m\| \le C_a$, so $\|(S^* - aI)x_m\| \to 0$. If $S - aI$ were Fredholm, then $S^* - aI$ would have closed range and finite-dimensional kernel. It would be bounded below modulo that kernel. But $x_m \rightharpoonup 0$, so its projection onto the finite-dimensional kernel tends to zero, contradicting the displayed singular-sequence estimate. Therefore $a \in \sigma_e(S)$ and

$$0 \in \sigma_e(aI - S) = \sigma_e(M_\psi).$$

The reciprocal has expansion

$$\frac{1}{a-z} = \sum_{n=0}^{\infty} a^{-(n+1)} z^n.$$

At the peak $P_m = N_m + L_m$, its squared-norm summand equals

$$\beta_{P_m}^2 a^{-2(P_m+1)} = \frac{2^{2L_m}}{a^{2(N_m+L_m+1)}}.$$

Since $L_m/(N_m+1) = N_m + 1 \to \infty$ and $a < 2$, these summands tend to infinity. Hence $1/\psi \notin B$.

Finally, the boundary cluster set of the affine function is exactly

$$\bigcap_{K \Subset \mathbb{D}} \overline{\psi(\mathbb{D} \setminus K)} = a - \mathbb{T}.$$

Indeed, radial sequences show that every $a - \zeta$, $|\zeta| = 1$, belongs to every set in the intersection. Conversely, using the compact disks $K = r\overline{\mathbb{D}}$ and then letting $r \uparrow 1$ leaves only that circle. Because $a = 3/2 > 1$, zero is not on it. Also $0 \notin \overline{\psi(\mathbb{D})} = a - \overline{\mathbb{D}}$, whereas $0 \in \sigma_e(M_\psi) \subset \sigma(M_\psi)$. This proves all assertions. $\square$

Verification and scope

The construction is entirely infinite-dimensional but uses only explicit scalar weights. The accompanying script checks the recursion, the divergence of the reciprocal peak terms, and the decay of the displayed singular-vector residual. These computations are sanity checks; the limiting estimates above are the proof.

The example satisfies more than the literal baseline assumptions: it is a separable Hilbert space, polynomials are dense, all point evaluations are bounded, and both the shift and its standard left inverse are bounded. It does not contradict positive spectral theorems for derivative-norm spaces, Muckenhoupt-weight Hardy spaces, or spaces whose multiplier algebra is known to be inverse-closed. Such results require additional structure absent from the source's general conjectures; see, for example, [3].

The source's Conjecture 1 is Zhu's separate weighted-Bergman reciprocal problem and is not refuted here. The counterexample addresses the source's general Conjectures 2 and 3 and its preceding general spectrum questions.

Bounded novelty check

The run registry, solution, attempt, and proof-gap indexes were searched for arXiv:2211.12236 and the core reciprocal, multiplier-spectrum, weighted-Hardy, and normalized-kernel phrases. The local 2000–present arXiv source corpus was searched for the same combinations. Exact web searches on July 21, 2026 used the arXiv id, full title, authors, "essential spectrum", "reciprocal problem", "bounded away from zero", and weighted-Hardy inverse-closedness. Publication and forward-reference records were also checked.

No paper explicitly answering the source's final questions or giving this counterexample was found. The later journal publication located in the search contains the reflexivity part of the preprint, not these final remarks. The closest mathematical source is Lindström–Miihkinen–Norrbo [3], which proves the desired spectrum formulas for important spaces under stronger structural hypotheses. Bourdon [2] treats general weighted Hardy spaces and weighted composition invertibility, but predates and does not address the present questions. Because irregular weighted-shift spectra are classical, expert review should still check for this precise application under alternative terminology.

Human review recommendation

Review as a likely valid full counterexample. The central checks are: the two-sided geometric recurrence for the singular vectors; the Fredholm contradiction from the orthonormal singular sequence; and the quantifier over all compact K in the boundary cluster set.

References

[1] G. Cao, L. He, and J. Li, *Evaluation functions and composition operators on Banach spaces of holomorphic functions*, arXiv:2211.12236 (2022).

[2] P. S. Bourdon, *Invertible weighted composition operators*, arXiv:1211.4190 (2012).

[3] M. Lindström, S. Miihkinen, and D. Norrbo, *Unified approach to spectral properties of multipliers*, arXiv:2002.07035 (2020).